

RAG TESTING

Checking whether the RAG system retrieves relevant contextual chunks correctly from the uploaded Microsoft report.

The retrieval testing was performed on:

- Azure Migrate Improvements
- Power Fx UDTs
- Microsoft Copilot Studio Improvements
- Power Platform Governance

RETRIEVAL ACCURACY FUNCTIONAL TESTING

Verify:

- semantic retrieval correctness
- FAISS ranking accuracy
- relevant chunk retrieval
- grounded contextual retrieval

Query	Expected Result	Actual Result	Pass/Fail
Azure Migrate Improvements	Migration workflow chunks retrieved	Correct chunks retrieved	Pass
Power Fx UDTs	Power Fx context retrieved	Correct chunks retrieved	Pass
Microsoft Copilot Studio Improvements	Agentic workflow chunks retrieved	Correct chunks retrieved	Pass
Power Platform Governance	Governance/security chunks retrieved	Correct chunks retrieved	Pass

Retrieval Accuracy Metrics

Total Queries Tested

4

Correct Retrievals

4

Final Retrieval Metrics

Metric	Result
Retrieval Accuracy	100%
Semantic Retrieval Success	100%
Relevant Chunk Retrieval	100%
Incorrect Retrieval Rate	0%

Retrieval Testing Result

Observed Output:

```
Power Apps modernizes app exper

MATCH SCORE: 100.00%
RESULT: PASS
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FINAL RETRIEVAL REPORT
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Retrieval Accuracy: 100.00%
=====
```

Final Conclusion

The RAG retrieval system successfully demonstrated:

- ✓ semantic search correctness
- ✓ FAISS retrieval accuracy
- ✓ relevant contextual ranking
- ✓ grounded chunk retrieval
- ✓ context-aware retrieval generation

The retrieval pipeline consistently returned highly relevant document chunks for all tested queries.

HALLUCINATION TESTING

Checking whether the RAG system generates unsupported or fabricated information outside the uploaded Microsoft report context.

The hallucination testing was performed against unsupported/fake technical and business claims.

HALLUCINATION FUNCTIONAL TESTING

Verify:

- unsupported claim rejection
- grounded response generation
- source-aware answer synthesis
- hallucination prevention

Query	Expected Result	Actual Result	Pass/Fail
Secure Boot hacked in 2026	Unsupported info rejection	Grounded response generated	Pass
Azure Migrate to Mars datacenters	Unsupported info rejection	Grounded response generated	Pass

Blockchain billing for SaaS activation	Unsupported info rejection	Grounded response generated	Pass
Microsoft France removed VAT requirements	Unsupported info rejection	Grounded response generated	Pass
AGI-powered self-thinking SaaS products	Unsupported info rejection	Grounded response generated	Pass
Azure servers powered by fusion reactors	Unsupported info rejection	Grounded response generated	Pass
Microsoft France tax fraud activities	Unsupported info rejection	Grounded response generated	Pass
Power Platform reading brain signals	Unsupported info rejection	Grounded response generated	Pass
Quantum teleportation migration	Unsupported info rejection	Grounded response generated	Pass
Secure Boot exposing passwords	Unsupported info rejection	Grounded response generated	Pass

Hallucination Accuracy Metrics

Total Hallucination Queries Tested

10

Correctly Grounded Responses

10

Final Hallucination Metrics

Metric	Result
Hallucination Prevention Accuracy	100%
Grounded Response Accuracy	100%
Unsupported Claim Rejection Accuracy	100%
False Information Generation	0%
Hallucination Rate	0%

Hallucination Testing Result

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RESULT: GROUNDED RESPONSE
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FINAL HALLUCINATION REPORT
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Hallucination Rate: 0.00%
(venv) PS C:\Users\india\Desktop\Market_Scout_New-main>
```

Final Conclusion

The RAG system successfully demonstrated:

- ✓ hallucination prevention
- ✓ grounded response generation
- ✓ unsupported claim rejection
- ✓ source-aware answer synthesis
- ✓ semantic retrieval correctness
- ✓ safe fallback behavior

The system consistently rejected fabricated claims and generated responses strictly grounded in the uploaded Microsoft report context.